

# WindOps Copilot - Turbine Report

Turbine ID: WTG-002 | Correlation ID: corr-7723501befc6  
Report Generated: 2026-06-26T13:23:45.677649

TURBINE HEALTH STATUS

65.0% Health

## Live Sensor Telemetry

| Sensor Parameter      | Value  | Unit |
|-----------------------|--------|------|
| Wind Speed            | 14.50  | m/s  |
| Rotor Speed           | 18.00  | RPM  |
| Blade Temperature     | 29.00  | °C   |
| Generator Temperature | 61.50  | °C   |
| Vibration             | 0.29   | mm/s |
| Power Output          | 2.80   | MW   |
| Operational Status    | active |      |

## Active Fault Alerts

| Parameter | Value | Threshold | Severity |
|-----------|-------|-----------|----------|
| Vibration | 0.29  | 0.25      | CRITICAL |

## Diagnostic Root Cause Analysis

|                     |                                                                                                                                                                         |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Diagnosed Cause:    | Rotor Imbalance or Blade Misalignment                                                                                                                                   |
| Confidence Level:   | 75.0%                                                                                                                                                                   |
| Technical Analysis: | Vibrations are elevated at 0.29 mm/s without critical heat signatures. This typical profile points to aerodynamic asymmetry, blade icing, or rotor coupling imbalances. |

## Operational Action Recommendations

| Recommended Action            | Description                                                                                                                                            | Risk | Production Loss | Confidence |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----------------|------------|
| Emergency Controlled Shutdown | Trigger pitch feathers and rotor brake immediately due to critical vibration threshold breach (>0.25 mm/s). Protect structural integrity of the asset. | LOW  | severe          | 100%       |

|                                   |                                                                                                                          |      |          |     |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------|------|----------|-----|
| De-rate / Curtail<br>Power Output | Reduce generator speed. (NOT RECOMMENDED: safety thresholds are violated, structural damage is likely if kept spinning). | HIGH | moderate | 10% |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------|------|----------|-----|

*Note: This report is generated by an autonomous multi-agent system. final actions must be reviewed and authorized by a human operator.*